

Analysis and cooking status

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09/10/25

Pass1 status

workflow_name	jobs	success
rgd-dmra-pass1_CxC_2-18339x116	36705	12913 (35.18%)
rgd-dmra-pass1_CuSn_1-18561x33	14687	14657 (99.80%)
rgd-dmra-pass1_empty_nominal-18399x2	113	113 (100.00%)
rgd-dmra-pass1_empty_zero_field-18316	239	239 (100.00%)
rgd-dmra-pass1_CuSn_2-18347x294	70791	5000 (7.06%)
rgd-dmra-pass1_LD2_2-18305x144	40076	0 (0.00%)
rgd-dmra-pass1_LD2_1-18431x16	7702	7702 (100.00%)
rgd-dmra-pass1_CxC_1-18445x13	6873	6873 (100.00%)

- Path: /cache/clas12/rg-d/production/pass1/recon
- Total: 177186, Done: 47497, Running: 15022, Pending: 114667
- (Done/Total): 26.81%, (Running/Total): 8.48%, (Pending/Total): 64.72%

Pass1 status

workflow_name	jobs	success
rgd-dmra-pass1_CxC_2-18339x116	36705	12913 (35.18%)
rgd-dmra-pass1_CuSn_1-18561x33	<pre> ----- rgd-dmra-pass1_CuSn_1-18561x33 ----- total: 14657 / 14687 = 99.7957% decmrg: 9524 / 9524 = 100.0000% recon: 4770 / 4770 = 100.0000% ana: 324 / 327 = 99.0826% anamrg: 26 / 33 = 78.7879% anaclean: 13 / 33 = 39.3939% </pre>	
rgd-dmra-pass1_empty_nominal-1839		
rgd-dmra-pass1_empty_zero_field-183		
rgd-dmra-pass1_CuSn_2-18347x294		
rgd-dmra-pass1_LD2_2-18305x144		
rgd-dmra-pass1_LD2_1-18431x16		
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clas12-config: Simulation and reconstruction

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- Goal: Add the GCARDS and YAML files to clas12-config to be able to use OSG with RG-D:

CxC variation:

```
<!-- RG-D Nuclear-target Assembly; variation can be for the flag assembly: LD2CuSn, LD2CxC -->  
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/target" factory="TEXT" variation="rgd_fall2023_CxC"/>  
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/cad/" factory="CAD" variation="rgd_fall2023_CxC"/>
```

LD2 variation:

```
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/target" factory="TEXT" variation="rgd_fall2023_LD2"/>  
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/cad/" factory="CAD" variation="rgd_fall2023_LD2"/>
```

CuSn variation:

```
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/target" factory="TEXT" variation="rgd_fall2023_CuSn"/>  
<detector name="/work/clas12/ouillon/tmp/ct2/experiments/clas12/targets/cad/" factory="CAD" variation="rgd_fall2023_CuSn"/>
```

Work on GEMC/GCARD: Results

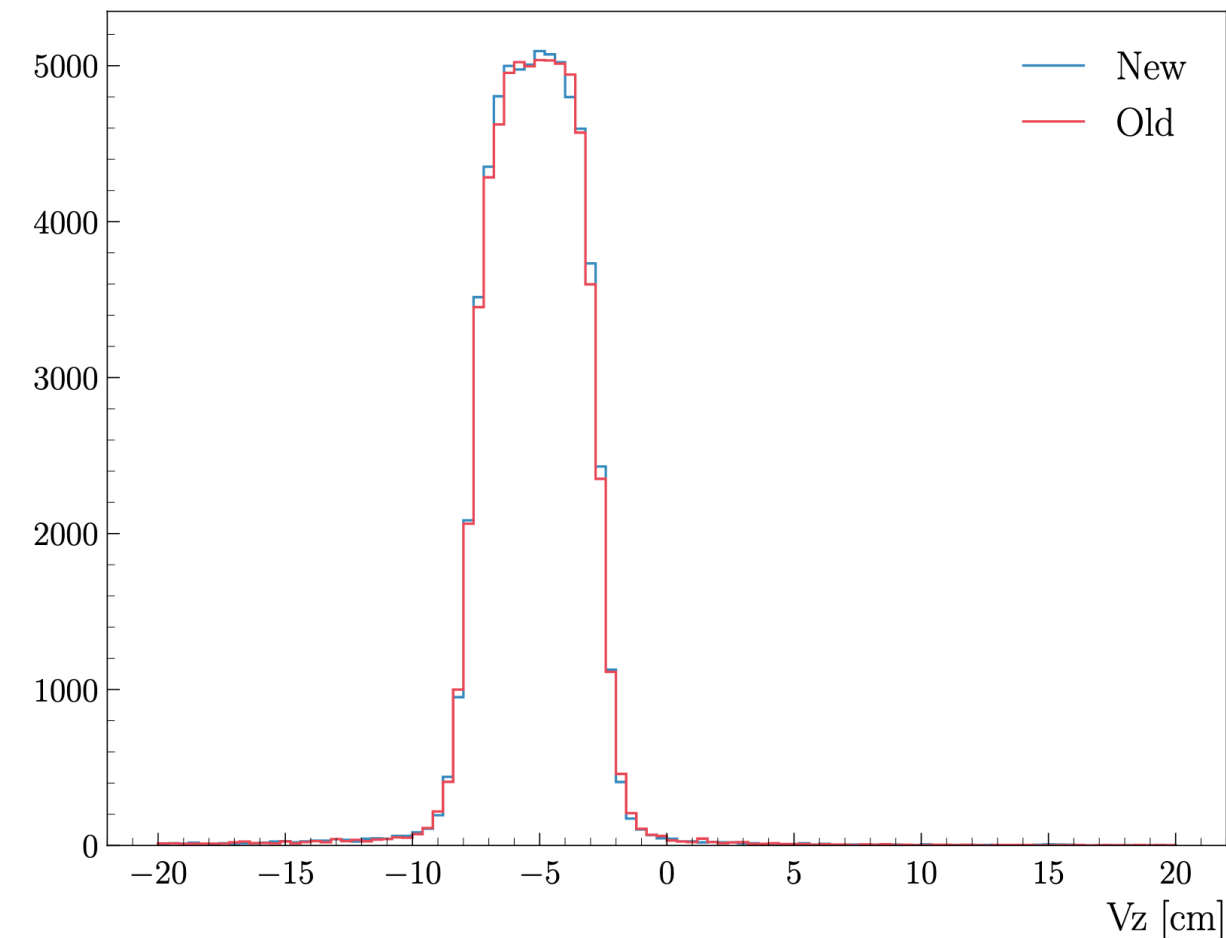
5

Old: GEMC 5.11 / CJ 11.1.0

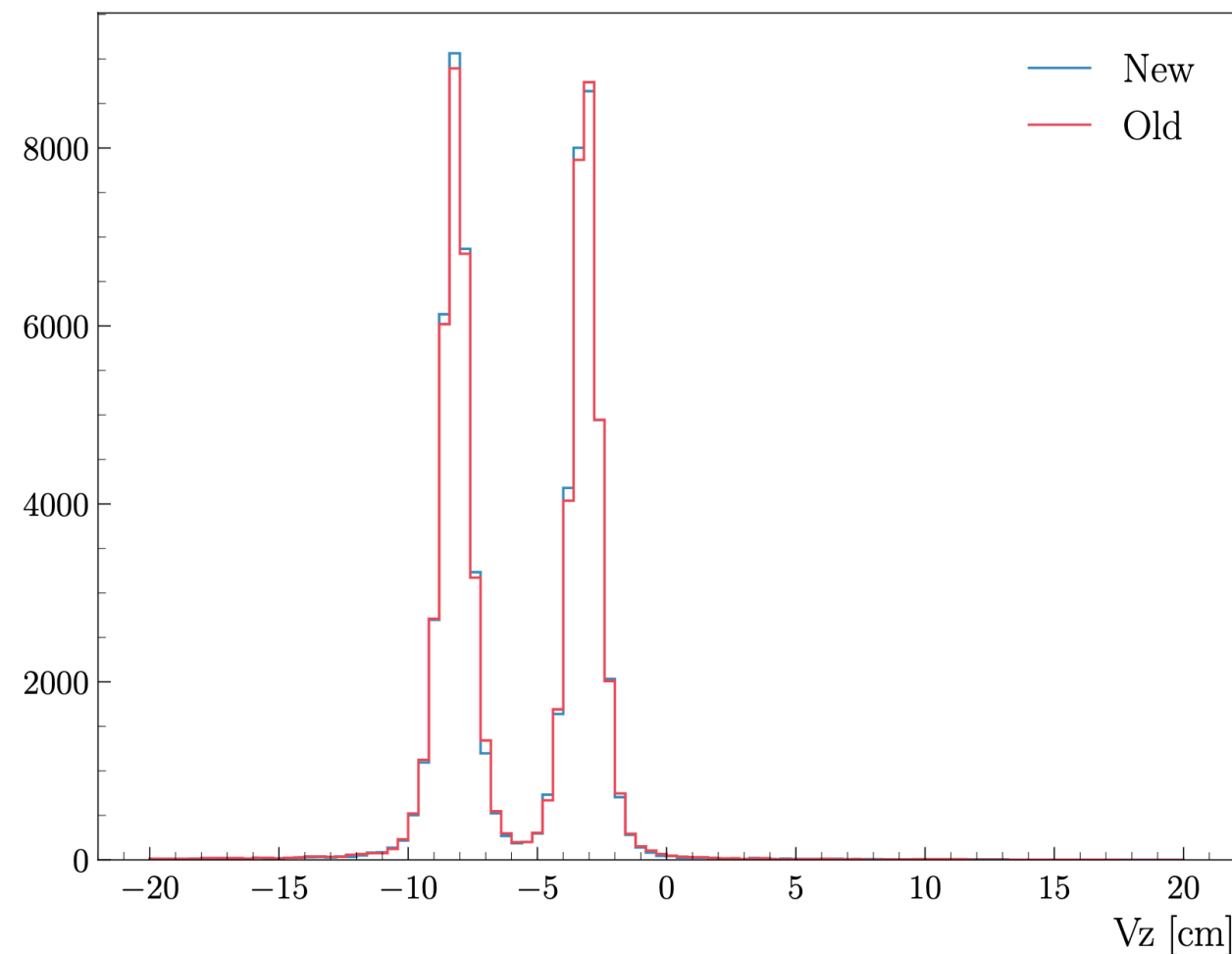
100 000 events in total (SIDIS event generator Daniel/Raphael)

New: GEMC dev / CJ 11.1.0

Electron ID2 Vz Distribution



Electron CxC Vz Distribution



Work on GEMC/GCARD: Results

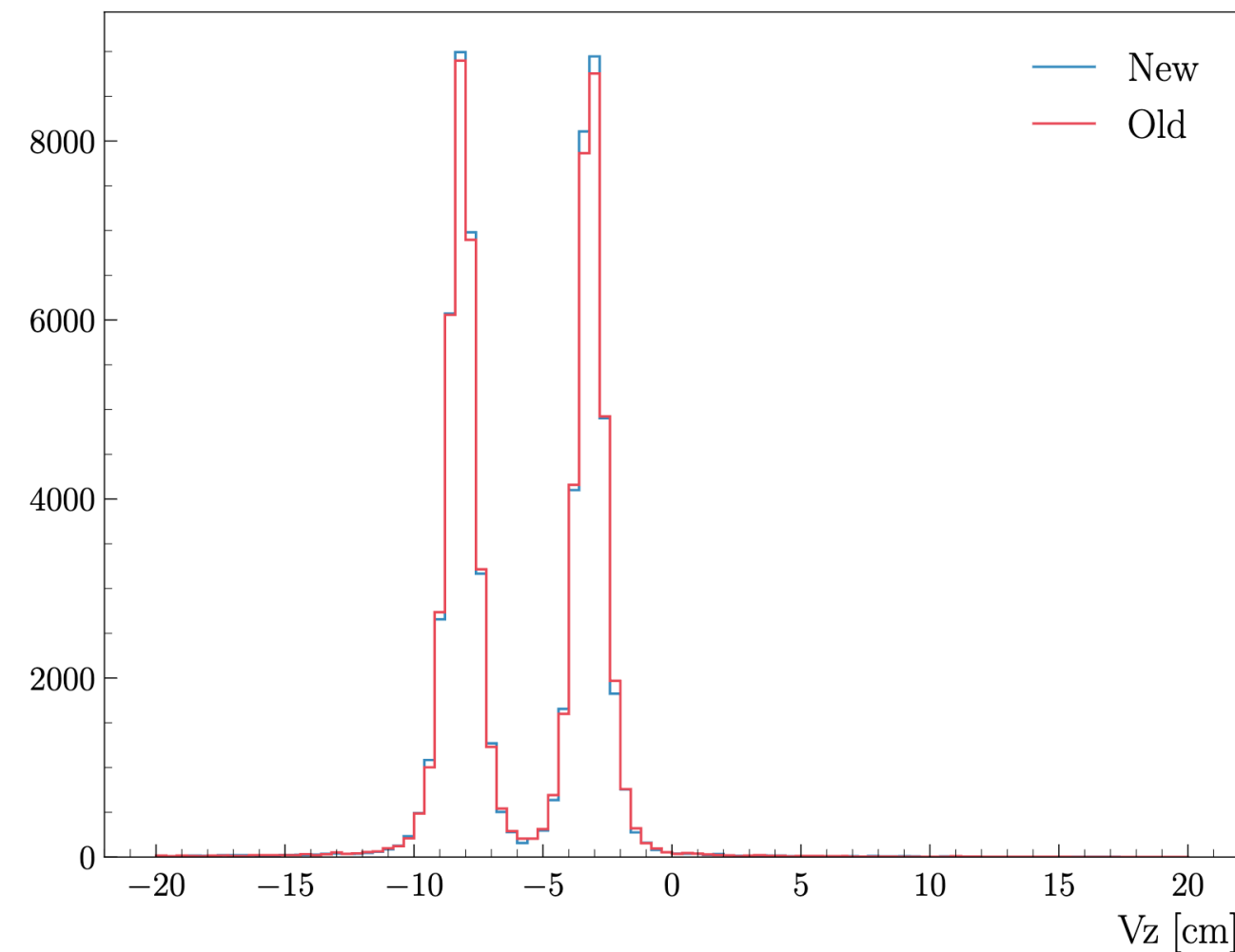
6

Old: GEMC 5.11 / CJ 11.1.0

100 000 events in total (SIDIS event generator Daniel/Raphael)

New: GEMC dev / CJ 11.1.0

Electron CuSn V_z Distribution



I don't see the first peak being small,
like in the data

Work on clas12-config:

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- I have worked to add the new gcard and yaml files to clas12-config:
 - Maurizo told me to add them to the dev branch, since we are using gemc/dev
 - This means that they will only be available for the dev/test version of the OSG portal until a new version of GEMC is released (which should happen soon).
 - In total, 4 GCARDS:
 - Available here: https://github.com/mathieuouillon/clas12-config/blob/dev/gemc/dev/rgd_fall2023_CxC.gcard
 - Standard CLAS12 configuration
 - Beamline OFF, outbending Full torus map
 - The target shift is in the geometry, not in the GCARD

 rgd_fall2023_Cu.gcard

 rgd_fall2023_CxC.gcard

 [rgd_fall2023_Sn.gcard](#)

 rgd_fall2023_ID2.gcard

Work on clas12-config:

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- I have worked to add the new gcard and yaml files to clas12-config:
 - For the YAML files, same as for the Pass1 cooking
 - + MC Engine:
 - `class: org.jlab.service.mc.TruthMatch`
`name: MC`
`dcMinistagger: "NOTONREFWIRE"`
`dcFeedthroughs: "OFF"`
 - + 2 DC option:
 - And mc variation: `variation: rgd_fall2023_mc`
 - 4 YAML files, one per GCARD with the same name.
 - Cu, Sn, and CxC are copies of the ID2 (symbolic link)
 -

 `rgd_fall2023_Cu.yaml`

 `rgd_fall2023_CxC.yaml`

 `rgd_fall2023_Sn.yaml`

 `rgd_fall2023_ID2.yaml`